- I²C-bus specification rev03 compatibility:
  - Controller and target modes
  - Multicontroller capability
  - Standard-mode (up to 100 kHz)
  - Fast-mode (up to 400 kHz)
  - Fast-mode Plus (up to 1 MHz)
  - 7-bit and 10-bit addressing mode
  - Multiple 7-bit target addresses (2 addresses, 1 with configurable mask)
  - All 7-bit addresses acknowledge mode
  - General call
  - Programmable setup and hold times
  - Easy-to-use event management
  - Clock stretching (optional)
- 1-byte buffer with DMA capability
- Programmable analog and digital noise filters
- SMBus specification rev 3.0 compatibility:
  - Hardware PEC (packet error checking) generation and verification with ACK control
  - Command and data acknowledge control
  - Address resolution protocol (ARP) support
  - Host and device support – SMBus alert
  - Timeouts and idle condition detection
- PMBus rev 1.3 standard compatibility
- Independent clock
- Wake-up from Stop mode on address match

Table 6. I2C implementation

Features(1)	I2C1
Standard-mode (up to 100 Kbit/s)	X
Fast mode (up to 400 Kbit/s)	X
Fast mode plus (Fm+) with 20 mA output drive I/Os (up to 1 Mbit/s)	X
Programmable analog and digital noise filters	X
SMBus/PMBus hardware support	X
Independent clock	X
Wake-up capability	X

1. X = supported.

3.24 Improved inter-integrated circuit interface (I3C)

The I3C interface handles communication between this device and others, such as sensors and the host processor, connected on an I3C bus.

An I3C bus is a two-wire, serial single-ended, multidrop bus, intended to improve a legacy I²C bus.

The I3C SDR-only peripheral implements all the features required by the MIPI® I3C specification v1.1. It can control all I3C bus-specific sequencing, protocol, arbitration, and timing, and can act as a controller (formerly known as master) or as a target (formerly known as slave). When acting as a controller, the I3C peripheral improves the features of the I2C interface while preserving some backward compatibility: it allows an I²C target to operate on an I3C bus in legacy I²C fast mode (Fm) or legacy I²C fast mode plus (Fm+), provided that the latter does not perform clock stretching. The I3C peripheral can be used with DMA to offload the CPU.

The I3C peripheral supports: